

## **Chapter 3 Study Notes**

### **The Cellular Level of Organization**

#### **LEARNING OBJECTIVES**

- Introduce the parts of a cell
  - Discuss the importance of the plasma membrane
  - Discuss the components of the cytoplasm
  - Compare and contrast mitosis and meiosis
  - Understand the effects aging has on the cell
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1. Compare and contrast an animal cell, plant cell, single celled prokaryote. How are they similar? Different?
  2. What are the three parts of an animal cell?
  3. In a separate document, create a two-column chart. In one column, place the name of the organelle (include all cell types) a picture of the organelle, what type of cell it's found in and what its main function is in the cell.
  4. What are the specific features of the plasma membrane that make it fluid and mosaic. Why is it considered to be: a lipid bilayer? amphipathic?
  5. What is meant when the plasma membrane is described as selectively permeable?
  6. Compare and contrast active and passive transport. What is facilitated diffusion? Osmosis? How does this differ from typical forms diffusion that occur regularly in the cell?
  7. What is diffusion influenced by?
  8. When does diffusion stop?
  9. Do materials still move across the membrane when equilibrium is met? Defend your answer.
  10. Give some examples of passive diffusion. Facilitated diffusion? What kinds of proteins are involved in facilitated diffusion?
  11. What is meant by primary active transport? Secondary active transport?
  12. Draw out the steps in receptor mediated endocytosis. Exocytosis?
  13. What are the three components in the cytoskeleton?
  14. Compare and contrast chromatin vs chromosomes.

15. What is meant by the endomembrane system? What organelles are responsible for this system. Trace the path of a secretory protein in this pathway.
16. What is meant by protein synthesis? What are the three major steps in protein synthesis (aka gene expression)?
17. What are the three sub -steps in transcription? Translation?
18. Create a chart with two columns. List the steps of gene expression and what major events (proteins involved etc) that occur in each step.
19. List the roles of mRNA, tRNA, and rRNA in gene expression.
20. What is the cell cycle? Use a pie chart to demonstrate the relative BIG stages in a somatic cell life.
21. In a separate chart with three columns: list the stages of interphase and mitosis, include (a) diagram(s) of the cellular events of each stage, and summarize the major events in each stage
22. Perform the same thing (chart) as above for Meiosis.
23. Compare and contrast the events of mitosis!